

Ambient heat and early childhood development: a cross-national analysis

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Background: Increasing evidence suggests that climate change, along with its cascading impacts on ecosystems, societies, and communities, has significant effects on both physical and mental health. However, less is known about how exposure to excessive heat early in life may influence the development of foundational skills that shape lifelong developmental trajectories. This study examined the effects of ambient heat on early childhood development across six countries, using geographic and time-stamped data on child development and ambient temperature. **Methods:** Our primary outcome is the Early Childhood Development Index. We used linear probability models with geographic and seasonality fixed effects to account for baseline climatic conditions, as well as other individual and contextual covariates to address potential selection bias. The sample comprised 19,607 children aged three and four from Georgia, The Gambia, Madagascar, Malawi, Sierra Leone, and the State of Palestine, all participants in Multiple Indicators Cluster Surveys collected between 2017 and 2020. We merged these data with temperature data from the ERA5-Land Monthly Aggregated Climate Dataset, calculating the mean monthly maximum temperature children experienced from birth to interview. **Results:** We found that children exposed to average maximum temperatures above 32°C were less likely to be developmentally on track compared to those exposed to cooler temperatures, even after accounting for baseline average climatic conditions and other covariates. Domain-specific models indicate that these effects were most pronounced in literacy and numeracy skills. Subgroup analyses revealed that the negative impacts were particularly severe for children in economically disadvantaged households and urban areas, and for those lacking access to adequate water and sanitation. **Conclusions:** This study highlights the potential impact of excessive heat on early childhood development, emphasizing the need for policies and interventions that enhance preparedness, adaptation, and resilience to support human development in an rapidly warming world. **Keywords:** Heat exposure; early childhood development; climate change; temperature.

Statement of relevance

The world recorded unprecedented temperatures in 2024, and a warming planet poses significant risks to both human development and health. While much is known about the broader impacts of climate change, less attention has been given to how high temperatures affect young children's development. Our study explored the effects of ambient heat on key developmental domains—literacy-numeracy, physical health, social-emotional well-being, and approaches to learning—across six countries: Georgia, The Gambia, Madagascar, Malawi, Sierra Leone, and the State of Palestine. We found that children exposed to average maximum temperatures above 32°C were less likely to be developmentally on track compared to those exposed to cooler temperatures. Moreover, these negative effects were most pronounced for children from economically disadvantaged backgrounds, those living in rural areas, and those lacking access to clean water and adequate sanitation. These findings highlight the urgent need for policies and interventions to protect human development in an increasingly warmer world.

Introduction

Climate change is profoundly changing the ecology of human development high (Cuartas & Morris-Perez, 2025). The year 2024 has been the warmest one on record, exceeding for the first time 1.5°C above pre-industrial levels for the annual average temperature set as a benchmark in the Paris Agreement (Romanello et al., 2024). Climate change can increase the risk for severe storms, atypical

precipitation patterns, loss of biodiversity, ocean acidification, increased pollution, and food and water scarcity, which altogether can in turn affect human societies and communities (Pörtner et al., 2022).

A warmer earth and its cascading effects on the physical environment, societies, and communities can undermine human health, learning, and well-being. Meta-analyses indicate that exposure to excessive ambient heat is associated with an increased risk for multiple physical and mental health problems in childhood and adulthood (Blom,

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Ortiz-Bobea, & Hoddinott, 2022; Thompson et al., 2023; Weeda, Bradshaw, Judge, Saraswati, & Le Souëf, 2024). Excessive heat can also inhibit children's and adolescents' learning and achievement in standardized tests (Park, Behrer, & Goodman, 2021; Prentice, Vergunst, Minor, & Berry, 2024). Moreover, hotter temperatures have been linked to an increased risk of interpersonal violence, including violence against children and gender-based violence (Evans, Gazze, & Schaller, 2025; Mahendran, Xu, Li, & Guo, 2021).

However, there is little evidence on how exposure to excessive ambient heat early in life, which is a sensitive period of brain development, may affect the development of foundational skills (Black et al., 2017; Cuartas et al., 2024; Draper et al., 2024; Herbst, 2024). High ambient temperatures may affect early childhood development by causing dehydration, neuroinflammation, heightened activation of the stress response system, and disrupted sleep (Berger et al., 2023; Sorensen & Hess, 2022; Walter & Carraretto, 2016; White et al., 2007; Yan, Liu, Wu, Jiang, & Zhuo, 2023). Young children are particularly vulnerable to ambient temperatures given their heightened sensitivity to environmental input, their underdeveloped biological systems, which are less capable of regulating and releasing heat via sweating, and their dependence on adults to seek water or cooler environments (Early Childhood Scientific Council on Equity and the Environment, 2023).

Excessive ambient heat can also influence young children's development through broader ecological mechanisms. For example, extreme heat can damage crops, increase the risk of food contamination, and promote the spread of disease vectors such as mosquitoes, all of which threaten children's health and nutrition, essential foundations for healthy development (Draper et al., 2024; Kemp et al., 2022; Schlenker & Roberts, 2009). Additionally, the well-documented effects of heat on mental health may impair parenting and reduce the quality of caregiver-child interactions, both of which are critical for the development of cognitive and socio-emotional skills (Bornstein, 2015; Cuartas et al., 2024). Excessive heat may also limit children's exposure to learning opportunities by making outdoor environments less safe and contributing to more sedentary behavior (Koepp, Lanza, Byrd-Williams, Bryan, & Gershoff, 2023). Consequently, understanding the developmental effects of early exposure to heat remains critical, especially considering that over one-third of children around the world are highly exposed to heatwaves and almost every child on earth is already experiencing the downstream consequences of climate change (Rees, 2021).

Understanding the developmental consequences of exposure to excessive ambient heat is challenging

due to issues of selection bias (Bailey et al., 2024). Specifically, individuals with certain characteristics “self-select” to live in different settings, and some of these characteristics may predict developmental outcomes, confounding the estimated effects. Similarly, comparing children living in hot and cold regions is not informative to isolate the effects of heat on child development, as hot and cold regions may differ in factors other than temperature (e.g., ecosystemic, economic, and social) that also influence development. Multiple confounders are difficult (if not impossible) to measure and hence unobserved by researchers, exacerbating selection bias issues.

This study addresses these gaps in the literature by studying the effects of excessive ambient heat during early childhood on early childhood development skills in six countries. Following work in environmental economics (e.g., Deschenes, 2014), we mitigate concerns about selection bias by linking geocoded data on children with high-resolution longitudinal temperature data. Specifically, our approach allows for comparisons among children living within the same subnational region, thereby reducing concerns about systematic differences in climatic, economic, or social conditions across regions, while also accounting for seasonality effects.

Furthermore, we examined whether the effects of ambient heat on child development varied between wealth groups, urban versus rural settings, and households with and without access to adequate water and sanitation. These factors were selected as potential moderators because they may either mitigate or exacerbate the risks associated with excessive heat (Balza, Cuartas, Gomez-Parra, & Serebrisky, 2025; Romanello et al., 2024)—for example, by providing resources to cope with heat (such as access to clean water) or by amplifying exposure and related risks (such as increased heat concentration in urban areas—known as the urban heat island effect—or contamination in households lacking adequate sanitation). Finally, given the unequal distribution of vulnerability to and gender-differentiated impact of different climate hazards (Awiti, 2022), we conducted an exploratory analysis to assess whether the effects of ambient heat differed between boys and girls.

Open practices statement

This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines. The analyses presented in this article were not formally preregistered. Neither the data nor the materials have been archived in a permanent third-party repository. The MICS and climate data used in this study are publicly available. Geocoded information from the MICS can be obtained upon request from the survey developers (mics.unicef.org).

Methods

Sample and procedures

We spatially combined data from the MICS and the ERA5-Land Monthly Aggregated Climate Dataset. Developed by UNICEF (2005), the MICS are nationally representative household surveys that employ stratified, multistage cluster sampling approaches and aim to collect internationally comparable data on children and women. Households are selected using probability proportional to size sampling within clusters, which correspond to the primary sampling units and typically include 20–25 geographically proximate households, often within a single census enumeration area. In this study, we used data from six countries included in the sixth round of the MICS: Georgia (2018), The Gambia (2018), the State of Palestine (2019–20), Madagascar (2018), Malawi (2019–20), and Sierra Leone (2017). We selected these countries because they were the only ones with available data on both geolocated household clusters, that is, geographic coordinates for each cluster, and child development, as measured by the Early Childhood Development Index (ECDI). These clusters serve as the primary units of analysis for spatial linkage with climate data.

Climate data were sourced from the ERA5-Land Monthly Aggregated Climate Dataset, produced by the European Centre for Medium-Range Weather Forecasts (Hersbach et al., 2020). This dataset combines satellite observations, in situ measurements, and numerical weather prediction models to generate consistent, continuous, and high-resolution climate records at a spatial resolution of approximately 9 km (0.1°). The data include hourly information on surface variables, including data on temperature and precipitation, from 1950 to approximately 2 months before the present. We overlaid the geolocated MICS clusters with the ERA5-Land climate grids in Google Earth Engine (see Figure 1). We conducted a complete case analysis, and the analytic sample for this study included 19,607 children aged three and 4 years in 3,646 unique clusters. Table 1 presents descriptive statistics of this sample.

Measures

Child development. We used the ECDI (Loizillon, Petrowski, Britto, & Cappa, 2017) to assess children's early development. The ECDI is a 10-item parent-reported scale that measures basic developmental milestones between the ages of 36 and 59 months in four domains: social-emotional, literacy-numeracy, approaches to learning, and physical. The ECDI is highly policy relevant because it is a proxy indicator for the Sustainable Development Goals Indicator 4.2.1 and one of the few comparable measures of child development widely used in large-scale surveys across countries. We computed a sum index of the 10 items and then characterized children to be developmentally "on-track" if they exhibited at least 50% of each domain's relevant skills, following the developer's guidelines (Loizillon et al., 2017). In the analytic sample, almost 63% of children were developmentally on track and 37% were not on track according to the ECDI.

Heat exposure. We leveraged the geographic coordinates collected by the MICS and exact data on children's birthdates and survey interview dates to establish heat exposure. After overlaying the MICS and ERA5 data spatially, we applied a buffer of 5 km around each MICS cluster point. The buffer accounts for local climate variability and random displacement of clusters, up to 2 km in urban areas and 5–10 km in rural areas, carried out by the MICS to ensure respondents' privacy. Within these buffers, we followed related literature (Aguilar-Gomez, Baquie, & Robyn, 2024; Deschenes, 2014) and extracted and averaged the climate variables for each buffer by month for each child, specifically calculating the mean

monthly maximum temperature that children were exposed to from birth to the interview date. Due to the absence of residential history data in MICS, we assign temperature exposure based on the child's current cluster of residence, implicitly assuming residential stability from birth through the time of the survey. Figure 2 presents the distribution of maximum temperature by country for the analytic sample.

Covariates. Our analysis included a set of child, caregiver, household, and contextual characteristics as covariates. The MICS data provided information on children's sex and age in months, maternal education (none, primary, secondary, or more); household wealth quintiles, as measured by a composite index of dwelling characteristics and household assets; and an indicator for urban versus rural areas. We also extracted a composite measure for dew point temperature, which is an absolute measure of atmospheric moisture, and surface pressure from the ERA5-Land as an additional atmospheric covariate. Our data also included information on children's exact birthdates, interview dates, and subnational regions (within countries) where clusters were located. Finally, we computed indicators for households' access (or lack thereof) to improved water and sanitation to assess potential heterogeneous effects of heat according to access to these basic services, following guidelines by the World Health Organization (2019).

Analytic approach

We used the linear probability model presented in Equation 1 to estimate the relation between heat exposure and child development. In this model, Y_{igt} represented the outcome variable for child i in geographic area g and date t . The model also included binary indicators for each degree Celsius, using exposure to average maximum temperatures below 26°C as the reference category (i.e., temperature bins). This threshold was informed by evidence suggesting potential adverse effects of temperatures above 26°C on mental health and behavior (Evans, 2019; Thompson et al., 2023) as well as the temperature distribution in the current sample. In this specification, $Tmax_{igt}$ represented the mean maximum temperature bins. A binned specification is common in the related literature and allowed us to account for potential non-linear associations between heat and child development (Aguilar-Gomez et al., 2024; Deschenes, 2014).

$$Y_{igt} = \sum_{b=1}^x \beta_j Tmax_{igt} + \sigma X_i + \gamma_g + \rho_t + \epsilon_{igt} \quad (1)$$

In the model, X_i was a set of covariates (children's age and sex, maternal education, wealth quartiles, area, dew point temperature and surface pressure, and child birthdate), and γ_g and ρ_t denoted subnational region-country fixed effects and month-year fixed effects, respectively. These fixed effects allowed us to control for seasonality effects and restrict comparisons between children within the same country and same subnational region who were exposed to different average maximum temperatures due to temporal variation in temperature levels relative to their birthdates and interview dates. Causal identification thus assumed that after controlling for baseline average climatic conditions (by including geographic and seasonality fixed effects), any residual fluctuations in temperature was unexpected or exogenous.

Furthermore, we estimated additional models using the indicators for being developmentally on track for each ECDI domain as the outcome variable to assess whether effects on a particular domain may drive the observed associations with the overall ECDI. We also conducted two sensitivity analyses to test the robustness of the results. First, although the selection of 26°C as the reference category in our binned specification was informed by prior evidence, we also conducted a

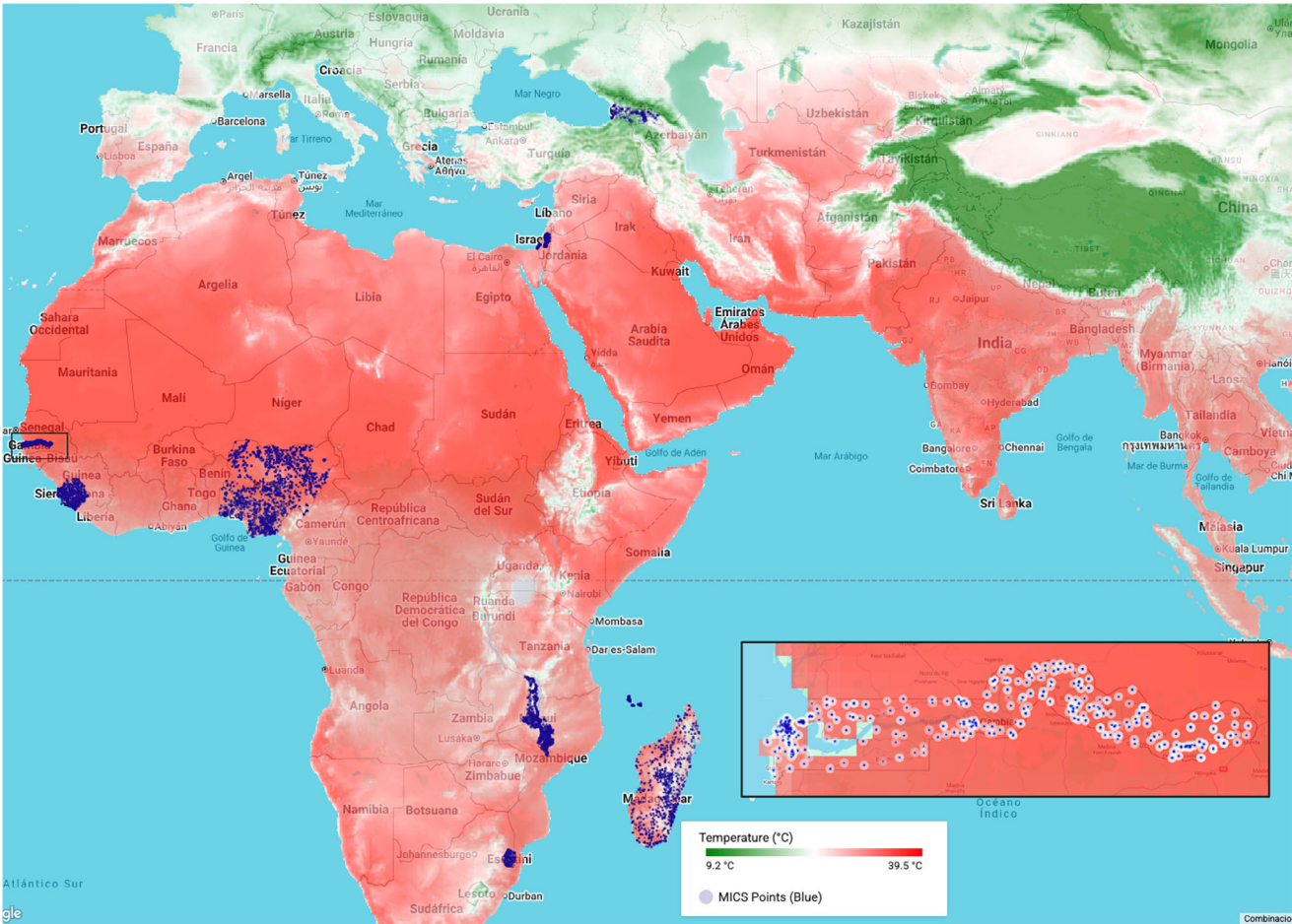


Figure 1 MICS clusters and maximum average temperature. Darker shades of red indicate higher temperatures. Dots represent MICS clusters (data were processed using Google Earth Engine; Gorelick et al., 2017)

Table 1 Descriptive statistics

Variable	%
Child development	
ECDI (on track)	62.9
Literacy–numeracy domain (on track)	14.7
Physical domain (on track)	94.4
Social–emotional domain (on track)	72.2
Approaches to learning (on track)	83.8
Child age	
3 years	57.2
4 years	42.8
Child sex	
Male	49.7
Female	50.3
Maternal education	
Less than primary	36.2
Less than secondary	40.0
More than secondary	23.8
Area	
Urban	28.7
Rural	71.3
N	19,607

data-driven binscatter least squares estimation as a sensitivity check. In this approach, the continuous temperature variable is divided into intervals (or bins), and each point in the plot represents the average ECDI score within a temperature bin,

plotted against the average temperature in that bin, allowing us to visualize the underlying functional relationship while smoothing out noise. Second, we estimated probit and logistic regression models instead of linear probability models, which account for the bounded nature of binary outcomes and allow for nonlinear relationships between the predictors and the probability of the outcome, providing a useful check on whether our findings are sensitive to the functional form of the model and distributional assumptions.

Finally, we assessed potential moderating effects or heterogeneity in the observed associations by estimating independent effects for different subgroups, including (1) wealth below and above the median, (2) urban versus rural, and (3) those with and without access to both improved water and improved sanitation. These analyses are helpful to understand the potential risk factors that may compound the effects of heat on early child development.

Results

Estimates of the effects of heat on early child development

Figure 3 summarizes results from our primary linear probability regression analysis estimating the association between temperature bins and being developmentally on track according to the ECDI. The Figure presents predicted margins (i.e., model-based predicted probabilities of the outcome at specific

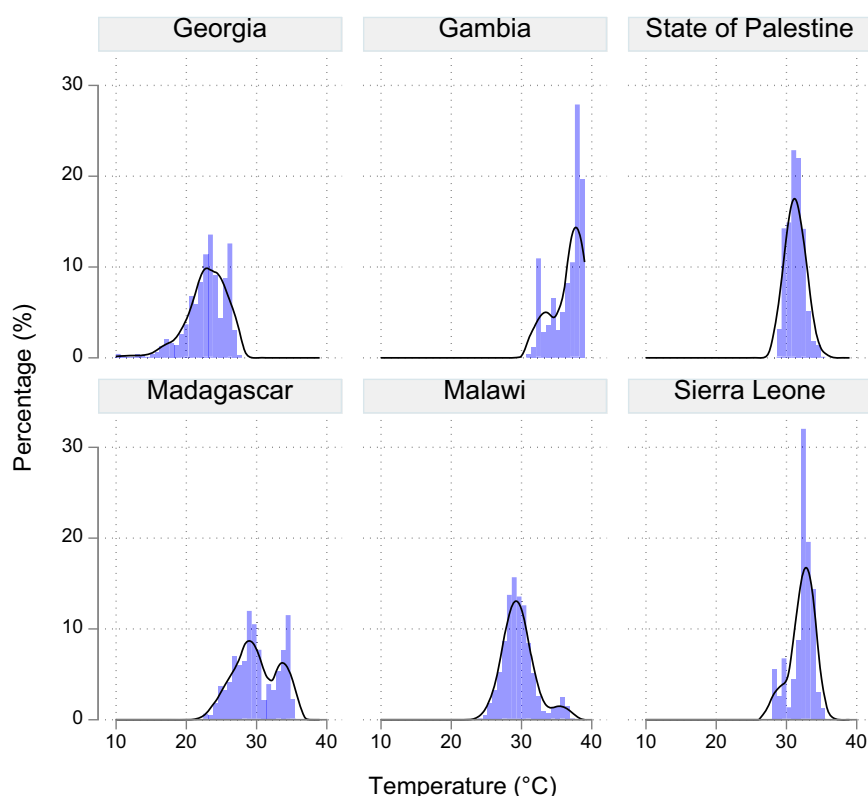


Figure 2 Monthly maximum temperature in sampled clusters, by country. This histogram displays the distribution of the average maximum temperature from birth to interview by country in the analytic sample. The figure presents an overlaid kernel density estimate for each country

values of the independent variable) for the unadjusted model which does not include any covariates, and for the model including all covariates and geographic and temporal fixed effects. The unadjusted estimates show a clear gradient, with children experiencing hotter average temperatures having a lower likelihood of being developmentally on track. For example, children who were exposed to an average maximum temperature of more than 30°C were between 14 ($SE = 0.019$) and 20.8 ($SE = 0.018$) percentage points less likely to be developmentally on track compared to those who were exposed to average maximum temperatures lower than 26°C. The adjusted model, including all covariates and our preferred model, showed that some, but not all, of these associations may be due to confounding. Specifically, it showed that children who were exposed to an average maximum temperature of 32°C were between 1.8 ($SE = 0.033$) and 7.7 ($SE = 0.039$) percentage points less likely to be developmentally on track compared to those who were exposed to average maximum temperatures lower than 26°C, with some bins not reaching statistical significance but being consistent in directionality. In general, both models showed the same drop in the probability of children being on track after exposure to average maximum temperatures higher than 30–31°C.

Table 2 presents results for the ECDI domain-specific analysis. Children who were exposed to monthly average maximum temperatures above 30°C had lower probabilities of being developmentally

on track in the literacy-numeracy domain, with a statistically significant association ranging between 5 ($SE = 0.025$) and 6.7 ($SE = 0.032$) percentage points, relative to those who were exposed to average maximum temperatures lower than 26°C. We also observed a statistically significant association between temperatures of 32°C and a 6.9 ($SE = 0.034$) percentage points lower likelihood of being developmentally on track in the socio-emotional domain, relative to temperatures below 26°C. Similarly, temperatures of 26°C are associated with a 2.2 ($SE = 0.013$) percentage point higher likelihood of being on track in the physical domain relative to temperatures below 26°C.

Figure 4 presents estimates from an alternative data-driven binscatter least squares estimation using continuous ECDI total scores and including all control variables and geographic and temporal fixed effects. The results were consistent with findings from our main specification, showing a negative association between ambient temperature and ECDI scores after accounting for baseline average climatic conditions and other individual and contextual covariates. Moreover, the predicted margins estimated using the probit and logit models were consistent with those from the linear probability model (see Figure S1).

Finally, while the main goal of this study was to assess the potential effects of excessive heat exposure from birth to the MICS interview date, extensive evidence also indicates that in-utero exposure to

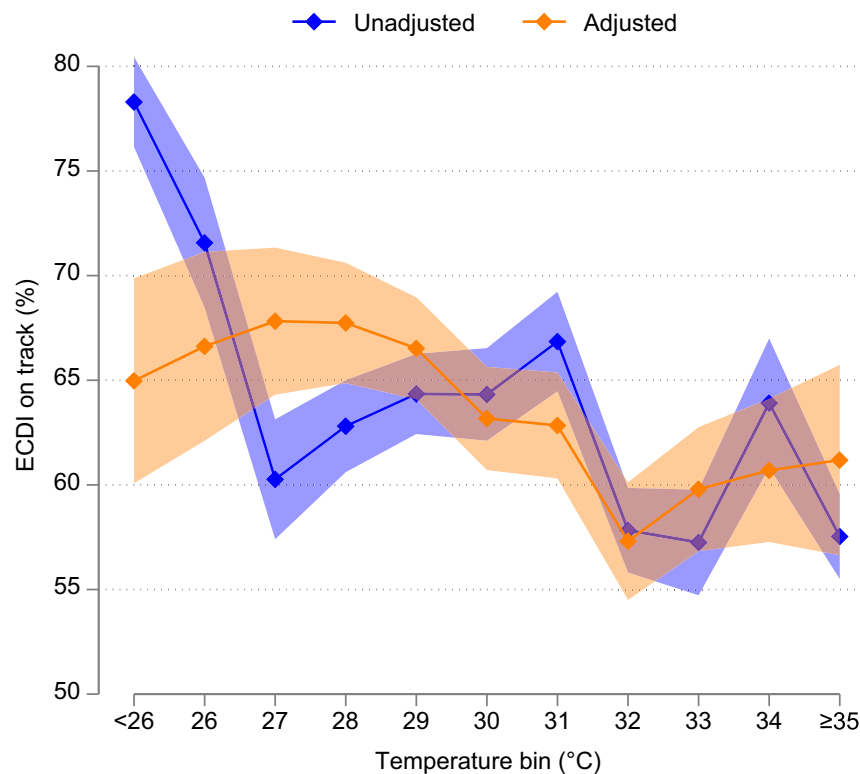


Figure 3 Association between temperature bins and the probability of being developmentally on track. The dots represent predicted margins, and the shaded region around the estimates represents 90% confidence intervals from linear probability models for the association between temperature bins and the probability of being developmentally on track as measured by the ECDI. Adjusted model refers to a model adjusting for children's age and sex, maternal education, wealth quartiles, area, dew point temperature and surface pressure, and child birthdate and interview month–year fixed effects and subnational region–country fixed effects

excessive heat can negatively impact human development outcomes (Brink et al., 2024). Consequently, we conducted a sensitivity analysis to examine the association between prenatal exposure to excessive heat and being developmentally on track (ECDI), assuming an average pregnancy duration of 9 months. Our results, presented in Table S1, show somewhat inconsistent associations between temperature bins and the ECDI. We observed statistically significant associations between temperatures of 26 and 27°C in the first trimester of pregnancy and a 4.1 ($SE = 0.020$) and 3.3 ($SE = 0.018$) percentage points higher likelihood of being developmentally on track, as well as between temperatures of 33°C and a 3.5 percentage points ($SE = 0.020$) lower probability of ECDI on track, relative to temperatures below 26°C. Moreover, temperatures of 34°C in the third trimester were associated with a 4.2 lower likelihood ($SE = 0.020$) of being developmentally on track relative to temperatures below 26°C. These results align with recent evidence on how heat can exacerbate risks for neurodevelopment early and late in pregnancy (Lin et al., 2025).

Heterogeneity in the effects of heat exposure on early child development

Subgroup analyses revealed pronounced heterogeneity in the association between heat exposure and

early childhood development (Figure 5). Children from economically disadvantaged households appear to bear a disproportionately higher burden. For example, children below the median experienced a reduced likelihood of being developmentally on track as temperatures increased, whereas this trend is not evident for those at the top of the wealth distribution. In addition, area-based differences emerge, with children exposed to high ambient temperatures and living in urban areas experiencing between 21.9 ($SE = 0.091$) and 26.6 ($SE = 0.087$) percentage point reductions in the likelihood of being developmentally on track, whereas those in rural areas did not experience such effects. Finally, children in households lacking improved water sources experienced stronger reductions in their likelihood of being developmentally on track when exposed to high heat (up to 13.9 percentage points, $SE = 0.082$).

Discussion

Climate projections suggest that current actions (and inactions) by governments and societies could lead to a temperature increase of 2.1–3.9°C above pre-industrial levels (Climate Action Tracker, 2024; Kemp et al., 2022). A warmer world may trigger cascading effects, including rising sea levels, more

Table 2 Associations between temperature bins and ECDI developmental domains

Temperature bins (°C)	ECDI domains			
	Literacy–Numeracy	Physical	Socio-emotional	Approaches to learning
<26 (reference)				
26	0.013 (0.022)	0.022* (0.013)	−0.005 (0.024)	0.009 (0.022)
27	−0.005 (0.023)	0.027 (0.016)	0.006 (0.026)	0.029 (0.026)
28	0.027 (0.023)	0.013 (0.016)	−0.002 (0.026)	0.010 (0.027)
29	−0.014 (0.023)	0.020 (0.017)	0.001 (0.027)	0.019 (0.027)
30	−0.050* (0.025)	0.027 (0.018)	0.002 (0.029)	−0.022 (0.029)
31	−0.023 (0.028)	0.025 (0.020)	−0.019 (0.032)	−0.024 (0.033)
32	−0.057* (0.029)	0.009 (0.021)	−0.069** (0.034)	−0.023 (0.034)
33	−0.066** (0.029)	0.031 (0.021)	−0.017 (0.035)	−0.039 (0.034)
34	−0.061** (0.030)	0.015 (0.022)	−0.020 (0.035)	−0.004 (0.035)
≥35	−0.067** (0.032)	0.020 (0.024)	−0.011 (0.040)	−0.020 (0.039)
N	19,607	19,607	19,607	19,607
Clusters	3,646	3,646	3,646	3,646

The table presents estimates from linear probability models, with corresponding country–cluster clustered standard errors in parentheses. All coefficients were estimated adjusting for children’s age and sex, maternal education, wealth quartiles, area, dew point temperature and surface pressure, and child birthdate and interview month–year fixed effects and subnational region–country fixed effects.

* $p < .05$; ** $p < .01$; *** $p < .001$.

frequent extreme weather events, the spread of vector-borne diseases, loss of ecosystem services, forced displacement, macroeconomic instability, conflict, resource shortages, and negative impacts on both physical and mental health (Kemp et al., 2022; Romanello et al., 2024). Despite the widespread recognition of these risks, including advocacy and discussion pieces highlighting the dangers of excessive heat for children (Early Childhood Scientific Council on Equity and the Environment, 2023; Rees, 2021), research linking ambient heat to early childhood development remains limited. This study leveraged novel spatial and temporal data to examine how excessive heat may impact the development of foundational skills in young children across six countries that vary in terms of socioeconomic characteristics, culture, and climatic conditions.

We found that children exposed to monthly average maximum temperatures above 32°C were less likely to be developmentally on track compared to those exposed to cooler temperatures. This association was consistent even after controlling for baseline average climatic conditions, by comparing children in the same subnational region and adjusting for seasonality effects, as well as other individual and contextual factors. These findings align with literature across multiple disciplines highlighting the environmental, health, and behavioral risks that

high temperatures pose to accelerate the proliferation of disease vectors and bacteria that cause gastrointestinal and respiratory illnesses (Kemp et al., 2022). Similarly, crop yields are adversely affected at temperatures above 34°C, and warmer conditions also heighten the risk of food contamination, which in turn threatens children’s health and nutrition and therefore development (Schlenker & Roberts, 2009). For example, one study estimated that a 2°C temperature increase could lead to a 7.4 percentage point rise in stunting, while temperatures exceeding 30°C are associated with a 2.2 percentage point increase in wasting rates (Blom et al., 2022). Finally, temperatures above 26°C can also increase adults’ mental health conditions, aggression, and interpersonal violence, with potential downstream effects on children’s safety and development (Evans, 2019; Thompson et al., 2023).

Domain-specific analyses suggest that the impact of heat on early child development may be particularly driven by its effects on cognitive literacy–numeracy skills, which aligns with prior research showing the adverse effects of heat on cognition. For example, temperatures above 32°C have been linked to a nearly 10% reduction in adult cognitive function and accelerated cognitive decline in the adult population (Choi, Lee, & Chang, 2023; Yin et al., 2024). Similarly, temperatures above 26.7°C can impair cognitive performance in children and

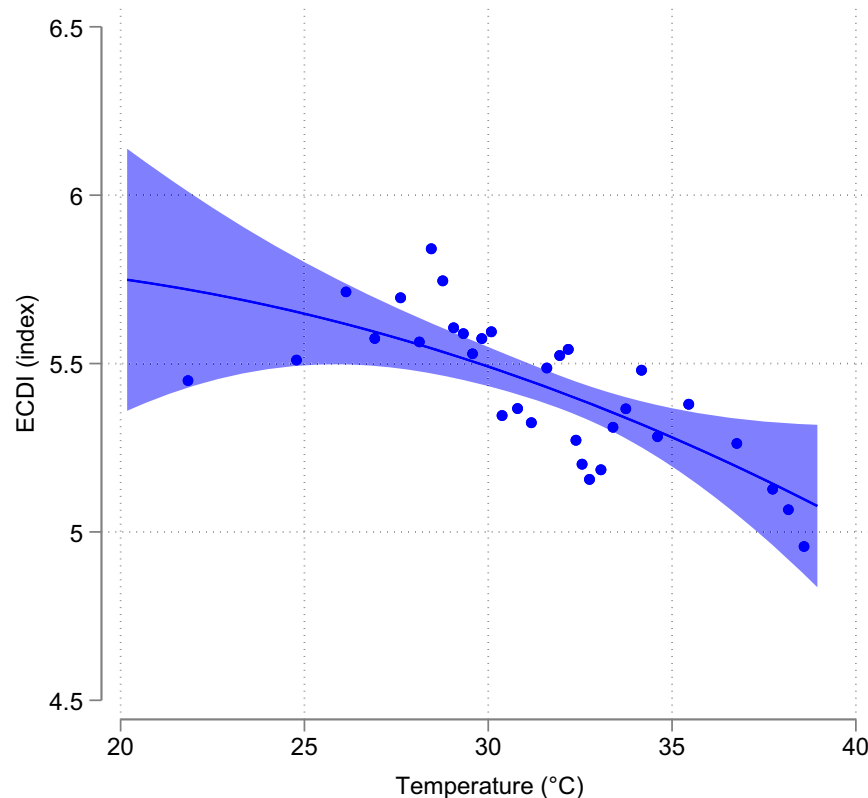


Figure 4 Association between ambient temperature and ECDI scores. This figure presents the predicted margins and fitted regression line from a binscatter analysis of the association between the average maximum temperature from birth to interview and the overall ECDI score. Each dot represents the average ECDI score for observations within a temperature bin, plotted against the average temperature in that bin. The model controls for children's age and sex, maternal education, wealth quartiles, area, dew point temperature and surface pressure, and child birthdate and interview month–year fixed effects and subnational region–country fixed effects

adolescents (Park, 2022; Park et al., 2021). Excessive heat can impact cognitive development directly (Piil et al., 2020) and indirectly through several mechanisms, including dehydration, increased activation of the stress response system, neuroinflammation, damage to nerve cells due to inadequate heat dissipation, and disrupted sleep (Berger et al., 2023; Sorensen & Hess, 2022; Walter & Carraretto, 2016; White et al., 2007; Yan et al., 2023). Similarly, it can reduce children's physical activity, increase sedentary behavior, and reduce opportunities for interaction with others outside the home (Koepp et al., 2023).

Conceptually, the same mechanisms linking excessive heat with cognitive outcomes could also affect children's physical and social–emotional development. However, in our sample, the evidence for such associations was less consistent. We found a statistically significant association between temperatures of 32°C and a lower likelihood of being developmentally on track in the socio-emotional domain, but not at higher temperatures. This inconsistent pattern may reflect measurement limitations in socio-emotional development relative to the literacy–numeracy domain, as noted in prior research (McCoy et al., 2016). Similarly, we observed a positive association between temperatures of 26°C and physical development, relative to cooler days,

but again, no consistent effects at higher temperatures. These mixed findings suggest caution in interpreting the effects of heat on physical and socio-emotional development in this sample. It is also possible that the effects of early-life heat exposure emerge more clearly in cognitive outcomes during early childhood, with impacts on other domains, such as socio-emotional functioning, becoming evident only later in life. Future studies using more precise measures and longitudinal designs are needed to clarify these patterns.

We also found that the effect of high temperatures on early child development is heterogeneous, with particularly severe effects on children from economically disadvantaged households, urban areas, and households with limited access to essential services. Families from economically disadvantaged backgrounds and in urban areas are more likely to live in housing and areas that trap heat and are less likely to have access to air conditioning, public cooling spaces, and clean drinking water, which increases the risk for heat stress and children's exposure to heat-related hazards (El Khayat et al., 2022; Gronlund, 2014). Furthermore, combined with inadequate access to water, which was found to be a significant moderator, heat exacerbates the risks of water scarcity, contamination, and disease transmission. Exposure to pollutants and

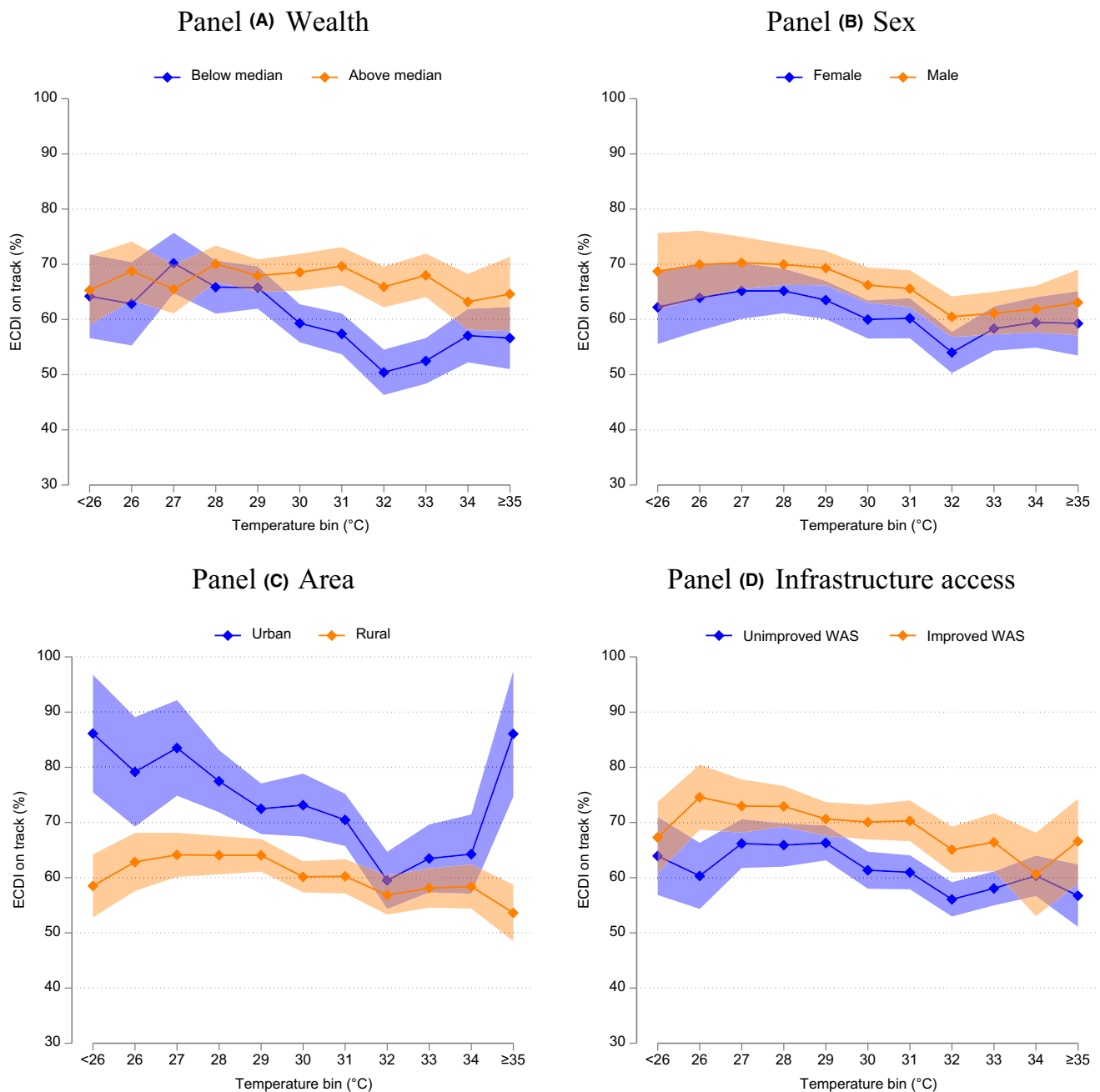


Figure 5 Predicted margins for the association between temperature bins and the probability of being developmentally on track by subgroups. Predicted margin results by subgroups from linear probability models adjusting for children's age and sex, maternal education, wealth quartiles, area, dew point temperature and surface pressure, and child birthdate and interview month-year fixed effects and subnational region-country fixed effects. WAS, water and sanitation

contamination during early childhood can disrupt synapse development in key brain areas, thereby impairing children's cognitive and social-emotional development (Walker et al., 2011). Conversely, access to adequate water can act as a protective buffer against the detrimental effects of heat on children by offering a readily available means of cooling (Balza et al., 2025).

Findings from the heterogeneity analyses highlight the urgent need to prioritize support for economically disadvantaged communities in efforts to mitigate heat-related risks to early childhood development. A

critical area of investment, as suggested by the current results, is the provision of reliable water and sanitation infrastructure, which can offer opportunities for cooling and reduce exposure to heat-related contaminants. Additional support may include cash transfer programs, which can enhance families' ability to access resources needed to cope with extreme heat, and the distribution of cooling tools, such as fans. Furthermore, the finding that heat effects are more pronounced in urban compared to rural areas, consistent with the urban heat island effect, underscores the importance of ecological and

infrastructure-based interventions, including increasing vegetation cover, implementing cool surface technologies, and enhancing urban infrastructure to support temperature regulation.

Limitations

This study makes significant contributions to the literature on the developmental consequences of excessive ambient heat, but it also has limitations. First, although we accounted for baseline average climatic conditions and other potential confounders, there may still be unobserved factors that could have biased the results. Additionally, while we can accurately assign temperature levels to children who remained in the same cluster throughout their lives, this is not possible for those who migrated far from their original MICS cluster. Therefore, future research should address selection biases, for example, by using longitudinal data with lagged dependent variables or individual fixed-effects models. Second, we operationalized heat exposure as the mean monthly maximum temperature, aligning with much of the existing literature that uses average temperatures to capture the cumulative intensity of heat exposure (Aguilar-Gomez et al., 2024; Deschenes, 2014). However, alternative approaches exist—for example, measuring the number of days exceeding a specific temperature threshold (e.g., Park et al., 2021), and there is currently no consensus in the field on the most appropriate way to conceptualize and operationalize heat exposure when studying its impacts on human health and development (Chloe et al., 2024).

Third, while the study draws on large-scale data from six diverse countries, the results may not be generalized to other settings. Future studies should therefore further investigate the effects of heat in additional contexts. Fourth, the ECDI used in this study is a parent-reported, limited-in-scope measure of child development with relatively low precision, which introduces uncertainty into the point estimates and coefficient attenuation. Additionally, heat exposure can make children tired, unmotivated, or sleepy, potentially affecting their performance on direct assessments of early childhood development. This raises concerns about endogeneity in the measurement of developmental outcomes. Future research should account for these nuances by employing multi-method assessment approaches that combine both direct assessments and parent reports to more reliably capture the developmental impacts of excessive heat. Finally, future research should explore potential mechanisms and additional moderators that could be relevant for interventions and policy, such as families' adaptive behaviors in response to heat.

Conclusion

Collectively, this study and prior research suggest that exposure to high temperatures can undermine children's early development, particularly foundational literacy and numeracy skills, with potential downstream effects on future learning and developmental trajectories. Children in economically disadvantaged households, urban areas, and those lacking access to clean water and adequate sanitation are at a heightened risk for maladaptive outcomes due to heat. These findings highlight the urgent need for further research, policies, and interventions that promote preparedness, adaptation, and resilience in a warming world. Additionally, targeted programs for economically disadvantaged families are essential to ensure the equitable protection of human development across the course of life.

Supporting information

Additional supporting information may be found online in the Supporting Information section at the end of the article:

Figure S1 Association between temperature bins and the probability of being developmentally on track: Probit and Logit models.

Table S1. Sensitivity check—marginal effects for the association between in-utero exposure to different temperature bins and the probability of being developmentally on track.

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Ethical considerations

Ethical approval is not applicable for this paper given that we used secondary data.

Data availability statement

The data that support the findings of this study are available from UNICEF. Restrictions apply to the availability of these data, which were used under license for this study. Data are available from <https://mics.unicef.org/> with the permission of UNICEF.

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Key points

What's known?

Climate change, along with its cascading impacts on ecosystems, societies, and communities, can affect human physical and mental health. However, little is known about how exposure to high ambient temperatures early in life might affect early childhood development

What's new?

This study used geographic and time-stamped information on children and ambient temperature in six countries to examine the effects of excessive heat on early childhood development

We found that children exposed to average maximum temperatures above 32°C were less likely to be developmentally on track compared to those exposed to cooler temperatures, even after controlling for baseline climate and other covariates

What's relevant?

These findings underscore the urgent need for policies and interventions to protect human development in an increasingly warmer world.

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